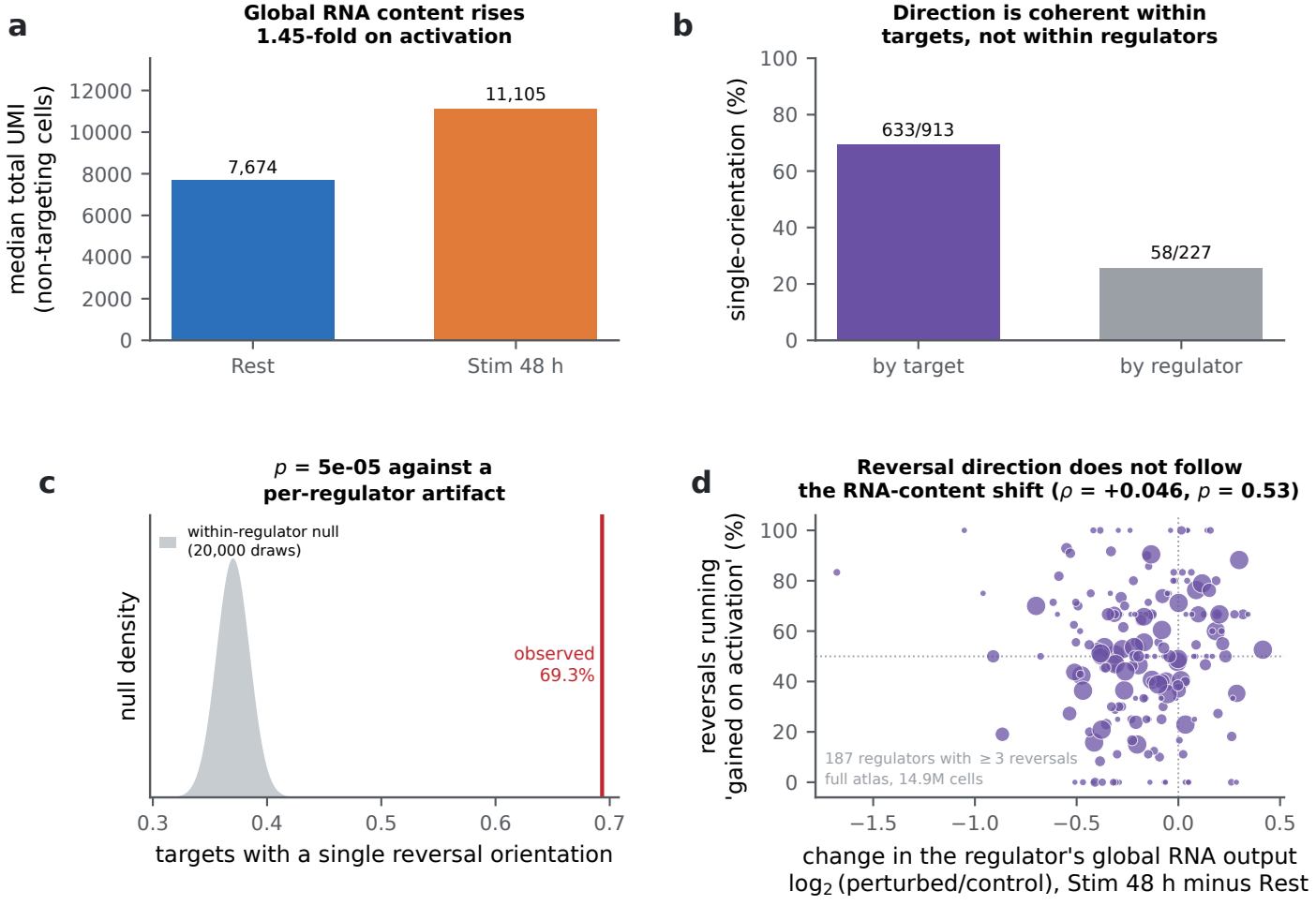

a
Global RNA content rises
1.45-fold on activation
median total UMI
(non-targeting cells)
12000
10000
8000
6000
4000
2000
0
7,674
11,105
Rest
Stim 48 h
b
Direction is coherent within
targets, not within regulators
single-orientation (%)
100
80
60
40
20
0
633/913
58/227
by target
by regulator
c
p = 5e-05 against a
per-regulator artifact
within-regulator null
(20,000 draws)
null density
observed
69.3%
0.3
0.4
0.5
0.6
0.7
targets with a single reversal orientation
d
Reversal direction does not follow
the RNA-content shift (ρ = +0.046, p = 0.53)
reversals running
'gained on activation' (%)
100
80
60
40
20
0
187 regulators with ≥ 3 reversals
full atlas, 14.9M cells
−1.5
−1.0
−0.5
0.0
0.5
change in the regulator's global RNA output
log2 (perturbed/control), Stim 48 h minus Rest